

MedGuide

Designing a Safer and Simpler Medication Decision Experience

Overview

Role

UX Designer | AI Product Designer | Frontend Developer

Duration

3months

Team Size

5 members

Tools

Figma, React Native, Canva, TensorFlow/PyTorch, MongoDB, Google Maps API

Project Type

Healthcare Mobile Application | AI-Assisted Decision Support

The Problem

When experiencing common illnesses such as colds, headaches, congestion, or seasonal allergies, many individuals struggle to select appropriate over-the-counter medications.

Pharmacy aisles contain dozens of seemingly similar products with varying ingredients, dosage instructions, side effects, and interactions.

Users often ask:

- Which medication is right for me?
- Am I taking the correct dosage?
- Will this interact with medications I already use?
- Which pharmacy nearby has it available?

The process requires significant research and medical literacy, creating confusion, frustration, and potential safety risks.

Research

Understanding User Pain Points

Research focused on common medication-related challenges faced by consumers.

Key Findings

Information Overload

Users encounter dozens of medication options with little guidance regarding differences.

Medication Adherence Issues

Many patients forget doses or fail to complete medication schedules.

Safety Concerns

Users frequently worry about side effects, allergies, and medication interactions.

Accessibility Challenges

Medical information is often difficult for non-medical users to understand.

Design Challenge

How might we help users make safer medication decisions without requiring extensive medical knowledge?

User Personas

Sarah – Busy Parent

Needs quick guidance when purchasing medication for common illnesses.

Pain Points:

- Limited time
- Confusing product labels
- Uncertainty about dosage

Michael – College Student

Frequently self-treats minor illnesses.

Pain Points:

- Limited healthcare access
- Difficulty understanding medication differences
- Forgetfulness with medication schedules

Linda – Older Adult

Manages multiple medications.

Pain Points:

- Concern about drug interactions
- Complex medication schedules
- Need for clear instructions

User Journey

Before MedGuide

Symptoms



Google Search



Multiple Health Websites



Pharmacy Visit



Compare Products



Uncertainty



Purchase Decision

After MedGuide

Symptoms



Guided Survey



AI Recommendation



Safety Verification



Medication Schedule



Pharmacy Availability



Confident Decision

Design Goals

The product was designed around five goals:

Simplicity

Reduce complexity during medication selection.

Trust

Increase confidence through safety-focused recommendations.

Accessibility

Present medical information using plain language.

Convenience

Combine medication selection, reminders, and pharmacy information in one experience.

Safety

Prevent harmful medication interactions whenever possible.

Information Architecture

Home



Symptom Survey



AI Recommendation



Medication Details



Drug Interaction Checker



Medication Calendar



Nearby Pharmacies

Design Process

Discovery

Analyzed common challenges users face when selecting over-the-counter medications.

Wireframing

Created low-fidelity user flows to simplify symptom entry and medication recommendations.

Prototyping

Built interactive Figma prototypes to test usability and navigation.

Iteration

Refined screens based on user feedback emphasizing clarity and trust.

Core Features

Symptom Survey

Users complete a guided questionnaire describing symptoms.

UX Goal

Reduce cognitive load by replacing open-ended searches with structured questions.

AI Recommendation Engine

The system analyzes symptoms and generates medication suggestions.

UX Goal

Provide confidence and reduce decision fatigue.

Drug Interaction Checker

The system verifies compatibility with existing medications and allergies.

UX Goal

Increase trust and improve user safety.

Medication Calendar

Users receive scheduled reminders and can track adherence.

UX Goal

Improve consistency and reduce missed doses.

Pharmacy Finder

Displays nearby pharmacies, pricing, and availability.

UX Goal

Reduce friction between recommendation and purchase.

Accessibility Considerations

Healthcare products must serve diverse populations.

Design decisions included:

- Large touch targets
- Plain-language instructions
- Clear information hierarchy
- High contrast interfaces
- Minimal medical jargon
- Consistent navigation patterns

Designing for Trust

Healthcare applications require a higher level of user trust than many consumer products.

Several principles guided design decisions:

Transparency

Recommendations are explained rather than presented as black-box outputs.

Safety First

Interaction checking was prioritized before recommendation delivery.

User Control

Users remain responsible for healthcare decisions.

Ethical AI

Recommendations assist decision-making rather than replace medical professionals.

Technical Constraints

Several constraints influenced the design process:

- AI prediction latency
- Accuracy requirements
- Medical information complexity
- Mobile device performance
- Real-time pharmacy data availability

The design balanced usability with technical feasibility.

Outcome

MedGuide successfully demonstrated an end-to-end healthcare experience combining:

- AI-assisted recommendations
- Medication adherence support
- Drug interaction safety checks
- Pharmacy discovery

The project transformed a fragmented medication-selection workflow into a unified, user-centered experience.

Reflection

The most important lesson from MedGuide was understanding how UX design changes when health and safety are involved.

Unlike traditional consumer applications, healthcare products require:

- Greater transparency
- Higher trust
- Strong accessibility
- Clear communication
- Safety-focused decision making

This project reinforced the importance of designing technology that empowers users while protecting their wellbeing.

Skills Demonstrated

- UX Research
- Healthcare UX
- Information Architecture
- User Journey Mapping
- Wireframing
- Prototyping
- Mobile UX Design
- AI Product Design
- Accessibility Design
- Human-Centered Design
- Ethical AI Design
- Healthcare Decision Support Systems

Potential User Risks

Risk	Mitigation
Incorrect symptom entry	Guided survey with validation
Misunderstanding recommendations	Plain-language explanations
Medication interactions	Drug Interaction Checker
Missed doses	Calendar reminders
Over-reliance on AI	Medical disclaimer + physician guidance

Error State Design

Healthcare applications must support users when things go wrong.

Several error scenarios were considered during the design process.

No Suitable Medication Found

Users receive guidance recommending consultation with a healthcare professional.

Allergy Conflict Detected

The system prevents recommendations and highlights the conflict.

Dangerous Drug Interaction Identified

A warning screen is displayed before any recommendation is presented.

Incomplete Survey Submission

Users are guided back to missing responses with clear instructions.

Pharmacy Inventory Unavailable

Alternative nearby pharmacies are suggested.

Designing for these situations improves both safety and user confidence.

Ethical AI Considerations

Because MedGuide incorporates artificial intelligence, ethical design principles were integrated throughout development.

AI as Assistance, Not Authority

The system assists users in decision-making but does not replace healthcare professionals.

Transparency

Users are informed that recommendations are generated using AI models trained on healthcare data.

Privacy

Health information is stored securely and used only to support the recommendation process.

Fairness

Future iterations should evaluate model performance across diverse demographic groups to minimize bias and improve reliability.

Healthcare AI should prioritize user wellbeing, safety, and informed decision-making above convenience alone.

Success Metrics

If deployed in a production healthcare environment, MedGuide could be evaluated using the following metrics:

- Symptom survey completion rate
- Recommendation acceptance rate
- Medication adherence rate
- Reminder engagement rate
- Drug interaction detection rate
- User confidence scores
- User satisfaction ratings
- Task completion time
- Reduction in medication selection uncertainty

These metrics would help determine whether MedGuide successfully improves both the healthcare experience and medication management outcomes.